9th Class • Punjab Board • PECTAA New Syllabus

Chapter 6: Mechanical Properties of Matter

Complete solved exercise — elasticity, Hooke's law, density, pressure, barometers, manometers and hydraulic systems.

Chapter 6 • Complete official exercise • 8 MCQs + 10 short + 10 constructed + 5 comprehensive + 12 numerical entries

What this chapter is about

• **Elasticity** is recovery of original shape after deforming forces are removed. Within the elastic limit, Hooke's law gives F = kx.

• **Density** ρ = m/V and **pressure** P = F⊥/A connect force, area and material distribution; 1 Pa = 1 N m^{-2}.

• **Liquid gauge pressure** p = ρgh depends on depth, density and g—not on container shape or total volume.

• **Pascal's law** transmits applied pressure through a confined fluid; hydraulic systems use $F_1/A_1 = F_2/A_2$.

Units used: 1 L = 10^{-3} m^3; 1 cm^3 = 10^{-6} m^3; 1 cm^2 = 10^{-4} m^2. Take g = 10 m s^{-2} and standard sea-level atmospheric pressure as 1.013×10^5 Pa.

A. Multiple Choice Questions

1. A wire is stretched by a weight w. If the diameter of the wire is reduced to half of its previous value, the extension will become:

(a) one-half (b) double
(c) one-fourth (d) four times

Answer: (d) four times — For the same wire material, load and length, ΔL ∝ 1/A ∝ 1/d^2. Halving diameter quarters the area, so extension becomes four times.

2. Four wires of the same material are stretched by the same load. Which elongates most? (a) L = 1 m, d = 1 mm (b) L = 2 m, d = 2 mm (c) L = 3 m, d = 3 mm (d) L = 4 m, d = 0.5 mm

(a) (a) (b) (b)
(c) (c) (d) (d)

Answer: (d) (d) — ΔL ∝ L/d^2. The ratios are 1, 0.5, 0.333 and 16 respectively, so the 4 m wire with 0.5 mm diameter extends most.

3. Two metal plates of area 2 and 3 square metres are placed in a liquid at the same depth. The ratio of pressures on the two plates is:

(a) 1:1 (b) √2:√3
(c) 2:3 (d) 4:9

Answer: (a) 1:1 — Fluid pressure at a given depth, p = ρgh, does not depend on the plate area. Force differs with area, but pressure is the same.

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4. The pressure at any point in a liquid is proportional to:

(a) density of the liquid
(b) depth below the liquid surface
(c) acceleration due to gravity
(d) all of the above

Answer: (d) all of the above — Gauge pressure $p = \rho gh$ increases directly with liquid density, depth and gravitational acceleration.

5. Pressure applied to an enclosed fluid is:

(a) increased and applied to every part
(b) diminished and transmitted to the walls
(c) increased in proportion to fluid mass and then transmitted
(d) transmitted unchanged to every portion of the fluid and walls of the containing vessel

Answer: (d) transmitted unchanged to every portion of the fluid and walls of the containing vessel — This is Pascal's law: an externally applied pressure change is transmitted undiminished throughout a confined fluid and to its container walls.

6. The principle of a hydraulic press is based on:

(a) Hooke's law
(b) Pascal's law
(c) conservation of energy
(d) conservation of momentum

Answer: (b) Pascal's law — Equal pressure on both pistons gives $F_1/A_1 = F_2/A_2$, allowing a large output force on the larger piston.

7. When a spring is compressed, what form of energy does it possess?

(a) Kinetic
(b) Potential
(c) Internal
(d) Heat

Answer: (b) Potential — Work done compressing the spring is stored as elastic potential energy and can be recovered during expansion.

8. What is the force exerted by the atmosphere on a rectangular block surface of length 50 cm and breadth 40 cm? Atmospheric pressure is 100 kPa.

(a) 20 kN
(b) 100 kN
(c) 200 kN
(d) 500 kN

Answer: (a) 20 kN — $A = 0.50 \times 0.40 = 0.20\ m^2$. $F = PA = 100{,}000 \times 0.20 = 20{,}000\ N = 20\ kN$.

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B. Short Answer Questions

6.1. Why do heavy animals like an elephant have a large area of the foot?

A large total foot area reduces pressure on the ground because P = W/A. For an elephant's very large weight, spreading the force over broad feet helps prevent excessive sinking and tissue or ground damage.

6.2. Why do animals like deer, which run fast, have a small area of the foot?

Small hooves concentrate weight on a smaller area, producing enough pressure to dig slightly into soft ground and provide grip for push-off. Small, light lower limbs also swing rapidly. Pressure alone is not the whole explanation of speed, but it describes the hoof–ground interaction intended here.

6.3. Why is it painful to walk bare-footed on pebbles?

Only small areas of the foot contact the sharp pebbles, so the same body weight produces large local pressure P = F/A. The high pressure on nerves and soft tissue causes pain; a flat surface spreads the force and reduces it.

6.4. State Pascal's law. Give an application of Pascal's law.

Pressure applied to any part of an enclosed fluid is transmitted undiminished to every part of the fluid and the containing walls. A **hydraulic brake** uses this principle: pedal pressure is transmitted through brake oil to wheel pistons, producing a larger braking force.

6.5. State what you mean by elasticity of a solid.

Elasticity is the property by which a solid returns to its original shape and size after the deforming force is removed, provided its elastic limit has not been exceeded. The work can be stored temporarily as elastic potential energy.

6.6. What is Hooke's law? Does an object remain elastic beyond the elastic limit? Give reason.

Within the elastic limit, extension or compression is directly proportional to applied force: F = kx. Beyond the elastic limit, the object does not return completely to its original dimensions because internal structure undergoes permanent rearrangement; it has undergone plastic deformation.

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6.7. Distinguish between force and pressure.

Force	Pressure
A push or pull; vector quantity.	Normal force per unit area; scalar quantity.
SI unit: newton (N).	SI unit: pascal (Pa) = N m^{-2}.
Depends on interaction, not surface area alone.	For the same force, decreases as contact area increases.
F = ma for a constant-mass net force.	P = F⊥/A.

6.8. What is the relationship between liquid pressure and depth?

For a liquid of fixed density at rest, gauge pressure is directly proportional to depth: p = ρgh. Doubling depth doubles pressure. Total pressure at depth also includes pressure on the surface, usually atmospheric pressure.

6.9. What is the basic principle used to measure atmospheric pressure by a simple mercury barometer?

The atmosphere pressing on mercury in a dish supports a vertical mercury column. With near-vacuum above the column, atmospheric pressure balances its hydrostatic pressure: P_atm = ρgh. At sea level, h is about 760 mm of mercury.

6.10. State the basic principle used in the hydraulic brake system of automobiles.

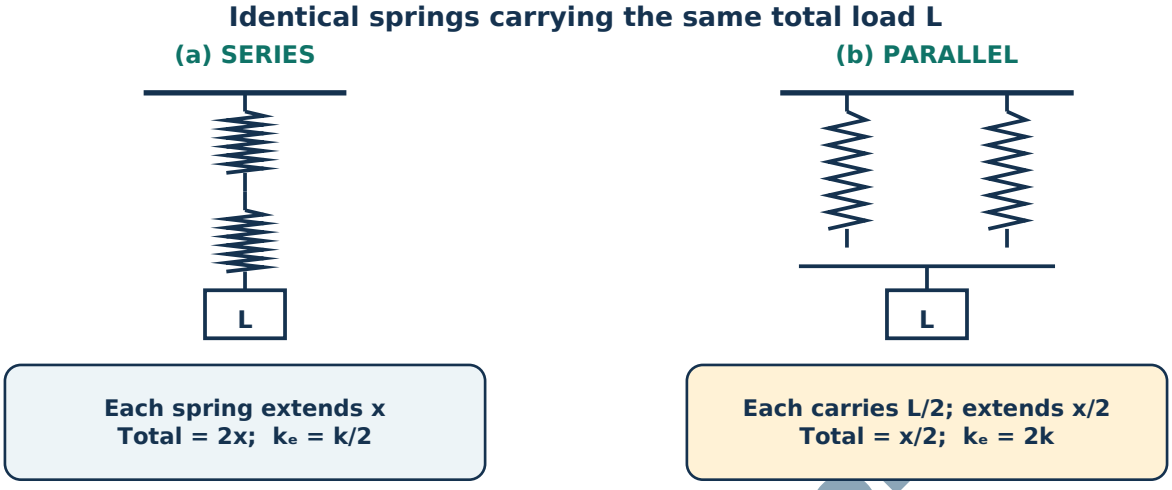
It uses **Pascal's law** in nearly incompressible brake fluid. Force on the small master piston creates pressure that is transmitted equally to larger wheel pistons. Their larger area gives greater force, pressing pads against discs or drums to slow the wheels by friction.

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C. Constructed Response Questions

6.1. A spring of constant k extends by x under load L. For two identical springs in (a) series and (b) parallel, determine total extension and effective spring constant.

Figure 1 — Series and parallel arrangements under the same load



Series: same force, extensions add. Parallel: same extension, forces add.

Arrangement	Force on each spring	Extension of each	Total extension	Effective constant
Series	L	x = L/k	2x	k/2
Parallel	L/2	(L/2)/k = x/2	x/2	2k

Series extensions add while the same force passes through both springs. Parallel springs have the same extension and their supporting forces add. All extensions remain within the elastic limit.

6.2. Springs are made of steel instead of iron. Why?

Spring steel has a high elastic limit, tensile strength and resilience: it can undergo repeated large elastic deflections and return to its original shape without taking a permanent set. Suitable alloys and heat treatment also improve fatigue resistance compared with ordinary iron.

6.3. Which material is more elastic? (a) Iron or rubber (b) Air or water

(a) **Iron** is more elastic in the physics sense because it requires much greater stress for the same strain and has a much larger Young's modulus; rubber is merely more stretchable. (b) **Water** is more elastic volumetrically because its bulk modulus is much larger, so it resists compression much more strongly than air.

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6.4. How does water pressure one metre below a swimming-pool surface compare with pressure one metre below the surface of a very large, deep lake?

The gauge pressures are essentially equal, $p = \rho gh$, if both contain water of the same density. Pressure depends on depth below the free surface, not on the pool's area, total volume or maximum depth. Atmospheric pressure adds equally to both for total pressure.

6.5. What happens to pressure in all parts of a confined liquid if pressure is increased in one part? Give a daily-life application.

By Pascal's law, the pressure increase is transmitted equally and undiminished to all parts of the liquid and container walls. In a hydraulic car lift or brake, pressure created at one piston produces force at every connected piston. Squeezing toothpaste also transmits pressure through the confined paste.

6.6. If air remains trapped above the mercury column of a barometer that should contain vacuum, how does it affect the column height?

The trapped air exerts downward pressure on the mercury. Then $P_{atm} = \rho gh + P_{air}$, so ρgh is less than atmospheric pressure and the column is **lower** than it should be. The barometer consequently reads too low and needs correction or refilling.

6.7. Why is the long neck not a problem for a giraffe when it raises its neck suddenly?

A giraffe's cardiovascular system regulates the hydrostatic pressure of its blood. A powerful heart and high arterial pressure supply the brain against gravity, while one-way venous valves, elastic pressure-buffering vessels and rapid vascular adjustments limit sudden pooling or pressure changes as the head rises. These adaptations maintain brain blood flow instead of allowing an uncontrolled ρgh change.

6.8. The end of the glass tube used in a simple barometer is not properly sealed; a leak is present. What will be its effect?

Air leaks into the space that should be vacuum and presses on the mercury, so the column falls and the reading becomes inaccurate. If the top becomes fully open to the atmosphere, pressure acts on both mercury surfaces and the useful height difference can collapse entirely.

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6.9. Comment on: “Density is a property of a material, not the property of an object made of that material.”

For a pure homogeneous material at fixed temperature and pressure, density ρ = m/V is intrinsic and does not depend on an object’s size or shape. However, an object’s **average density** can differ because of hollow spaces, pores or mixed materials. The statement is therefore correct for material density, with that distinction.

6.10. How is the load of a large structure estimated by an engineer?

The engineer calculates component volumes and uses W = mg = ρVg to estimate self-weight, then adds expected occupants, furniture, equipment and other live loads, together with relevant wind, snow or seismic effects. Loads are combined with safety factors and distributed to supports; bearing pressure is checked using P = F⊥/A so that soil and materials are not overstressed.

D. Comprehensive Questions

6.1. What is Hooke’s law? Give three applications of this law.

Hooke’s law: Within the elastic limit, the extension or compression of an elastic body is directly proportional to the applied force. For a spring, F = kx, where k is the spring constant in N m^{-1}.

1. Spring balance and force sensors: A calibrated extension measures weight or force through F = kx.

2. Vehicle suspension and shock absorbers: Springs deform to support loads and store energy temporarily, improving comfort and road contact.

3. Elastic supports and vibration control: Spring mountings in instruments, machines and furniture control deflection and reduce transmitted shocks.

The law is valid only below the elastic limit. Exceeding it causes permanent deformation, so a stretched spring may no longer follow the same calibration.

6.2. Describe the working and applications of a simple mercury barometer and a manometer.

Mercury barometer: Completely fill a closed glass tube with mercury and invert it in a mercury dish without admitting air. The column falls until atmospheric pressure on the dish balances ρgh, leaving a near-vacuum above. Measure the vertical height h from the dish surface; at sea level it is about 0.76 m.

Barometer: P_atm = ρ_Hggh

Applications include measuring atmospheric pressure, indicating weather changes and estimating altitude because pressure decreases with height.

U-tube manometer: Connect one arm to the gas supply and leave the other open to atmosphere. A level difference h shows the pressure difference. If gas pressure

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depresses its own side, it is greater than atmospheric; if its side rises, it is lower.

$$P_{gas} = P_{atm} + \rho gh$$

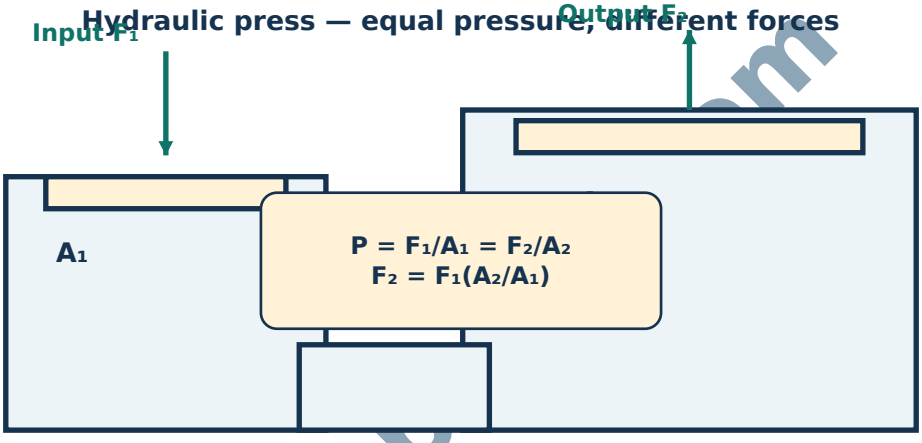
$$P_{gas} = P_{atm} - \rho gh$$

Manometers measure gas pressure in laboratories and supply systems. Use vertical level difference, not the length of liquid along a sloping tube.

6.3. Describe Pascal’s law. State its applications with examples.

Pascal’s law states that a pressure change applied to a confined fluid is transmitted equally and undiminished to every point of the fluid and to the container walls.

Figure 2 — Force multiplication in an ideal hydraulic press



If $A_2 > A_1$, output force is larger; the small piston moves through a greater distance.

Ideal work is conserved: $F_1 d_1 \approx F_2 d_2$ (neglecting losses).

$$P = F_1/A_1 = F_2/A_2$$

$$F_2 = F_1(A_2/A_1)$$

Hydraulic press: a small input force compresses material or lifts a heavy platform. **Hydraulic brakes:** the master piston transmits pressure to wheel pistons, which press pads against discs. **Vehicle lifts and jacks:** a small pump force raises cars for servicing.

Force multiplication is not energy creation. Ideally the small piston travels farther, so input work $F_1 d_1$ approximately equals output work $F_2 d_2$ minus losses.

6.4. On what factors does the pressure of a liquid in a container depend? How is it determined?

Consider a vertical liquid column of cross-sectional area A and height h. Its volume is Ah and mass is ρAh. Its weight is ρAhg, which produces force on the base.

$$P = F/A = (\rho Ahg)/A = \rho gh$$

Gauge pressure therefore depends directly on **liquid density ρ**, **depth h** and **gravitational acceleration g**. It is independent of base area, container shape and total liquid volume. At the same depth in a continuous static liquid, pressure is the same in every direction.

Total pressure = surface pressure + ρgh

In an open container: $P_{total} = P_{atm} + \rho gh$

Determine gauge pressure with ρgh or a manometer. Add atmospheric pressure only when absolute total pressure is requested.

6.5. Explain that the atmosphere exerts pressure. What are its applications? Give at least three examples.

Air has weight and its molecules continually collide with surfaces. The resulting normal force per unit area is atmospheric pressure, about 1.013×10^5 Pa at sea level. A sealed steam-filled can crumples when cooled because internal pressure falls and outside atmospheric pressure exceeds it.

• **Drinking straw:** Reducing pressure in the straw lets atmospheric pressure on the drink push liquid upward.

• **Syringe and medicine dropper:** Pulling a plunger or releasing a squeezed bulb lowers internal pressure and atmosphere pushes liquid in.

• **Suction cup:** Expelling air leaves lower pressure beneath the cup; outside atmospheric pressure presses it to a smooth surface.

• **Vacuum cleaner:** A fan creates a pressure difference and atmospheric pressure drives dust-laden air into the machine.

We normally do not feel the large pressure because fluids and air inside the body exert balancing pressure and it acts in all directions.

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E. Numerical Problems

6.1 A spring is stretched 20 mm by a load of 40 N. Calculate its spring constant. If an object causes an extension of 16 mm, what is its weight?

Given: $F_1 = 40$ N; $x_1 = 20$ mm $= 0.020$ m; $x_2 = 16$ mm $= 0.016$ m.

$k = F_1/x_1 = 40/0.020 = 2000 \text{ N m}^{-1} = 2 \text{ kN m}^{-1}$

$W_2 = kx_2 = 2000 \times 0.016 = 32$ N

Answer: $k = 2$ kN m^{-1}; weight = 32 N.

6.2 The mass of 5 litres of milk is 4.5 kg. Find its density in SI units.

Given: $m = 4.5$ kg; $V = 5$ L $= 5 \times 10^{-3}$ m^3.

$$\rho = m/V = 4.5/(5 \times 10^{-3}) = 900 \text{ kg m}^{-3}$$

Answer: $\rho = 0.9 \times 10^3$ kg m^{-3} = 900 kg m^{-3}.

6.3 A solid of mass 60 g raises water in a measuring cylinder from 40 cm³ to 44 cm³. Calculate its density.

Given: m = 60 g = 0.060 kg; V = 44 − 40 = 4 cm³ = 4×10^{-6} m³.

$$\rho = m/V = 0.060/(4 \times 10^{-6}) = 15{,}000 \text{ kg m}^{-3}$$

Answer: $\rho = 1.5 \times 10^4$ kg m^{-3} (equivalently 15 g cm^{-3}).

6.4 A block of density 8×10^3 kg m^{-3} has volume 60 cm³. Find its mass.

Given: $\rho = 8000$ kg m^{-3}; V = 60 cm³ = 60×10^{-6} m³.

$$m = \rho V = 8000 \times 60 \times 10^{-6} = 0.48 \text{ kg}$$

Answer: 0.48 kg = 480 g.

6.5 A brick measures 5 cm × 10 cm × 20 cm and has mass 5 kg. Calculate the maximum and minimum pressure it can exert on a horizontal surface.

Given: Weight = mg = 5 × 10 = 50 N. Maximum P uses smallest face; minimum P uses largest face.

$$\text{Smallest area} = 0.05 \times 0.10 = 0.005 \text{ m}^2$$

$$P_{max} = 50/0.005 = 1.0 \times 10^4 \text{ Pa}$$

$$\text{Largest area} = 0.10 \times 0.20 = 0.020 \text{ m}^2$$

$$P_{min} = 50/0.020 = 2.5 \times 10^3 \text{ Pa}$$

Answer: $P_{max} = 1 \times 10^4$ Pa; $P_{min} = 2.5 \times 10^3$ Pa.

6.6 What will be the barometer column height at sea level if mercury is replaced by water of density 1000 kg m^{-3}? Mercury density is 13.6×10^3 kg m^{-3}.

Given: Use $h_{Hg} = 0.760$ m and the same atmospheric pressure for both liquids.

$$\rho_w g h_w = \rho_{Hg} g h_{Hg}$$

$$h_w = (\rho_{Hg}/\rho_w) h_{Hg} = (13.6 \times 10^3/1000) \times 0.760 = 10.336 \text{ m}$$

Answer: Approximately 10.3 m of water.

6.7 In a hydraulic brake, 500 N acts normally on a 5 cm² piston. Find the pressure transferred to the oil and force on a second piston of area 20 cm².

Given: $F_1 = 500$ N; $A_1 = 5$ cm² $= 5 \times 10^{-4}$ m²; $A_2 = 20$ cm² $= 20 \times 10^{-4}$ m².

$$P = F_1/A_1 = 500/(5 \times 10^{-4}) = 1.0 \times 10^6 \text{ N m}^{-2}$$

$$F_2 = PA_2 = 10^6 \times 20 \times 10^{-4} = 2000 \text{ N}$$

Answer: Pressure = 1.0×10^6 Pa; output force = 2000 N.

6.8 Find the water pressure on a deep-sea diver at a depth of 10 m, where sea-water density is 1030 kg m⁻³.

Given: $\rho = 1030$ kg m⁻³; $h = 10$ m; $g = 10$ m s⁻².

$$p = \rho gh = 1030 \times 10 \times 10 = 103{,}000 \text{ N m}^{-2} = 1.03 \times 10^5 \text{ Pa}$$

Answer: Gauge water pressure = 1.03×10^5 Pa. Total including atmosphere would be about 2.04×10^5 Pa.

6.9 The small and large pistons of a hydraulic press have areas 10 cm² and 100 cm². What force on the small piston lifts a car of weight 4000 N?

Given: $A_1 = 10$ cm²; $A_2 = 100$ cm²; $F_2 = 4000$ N.

$$F_1/A_1 = F_2/A_2$$

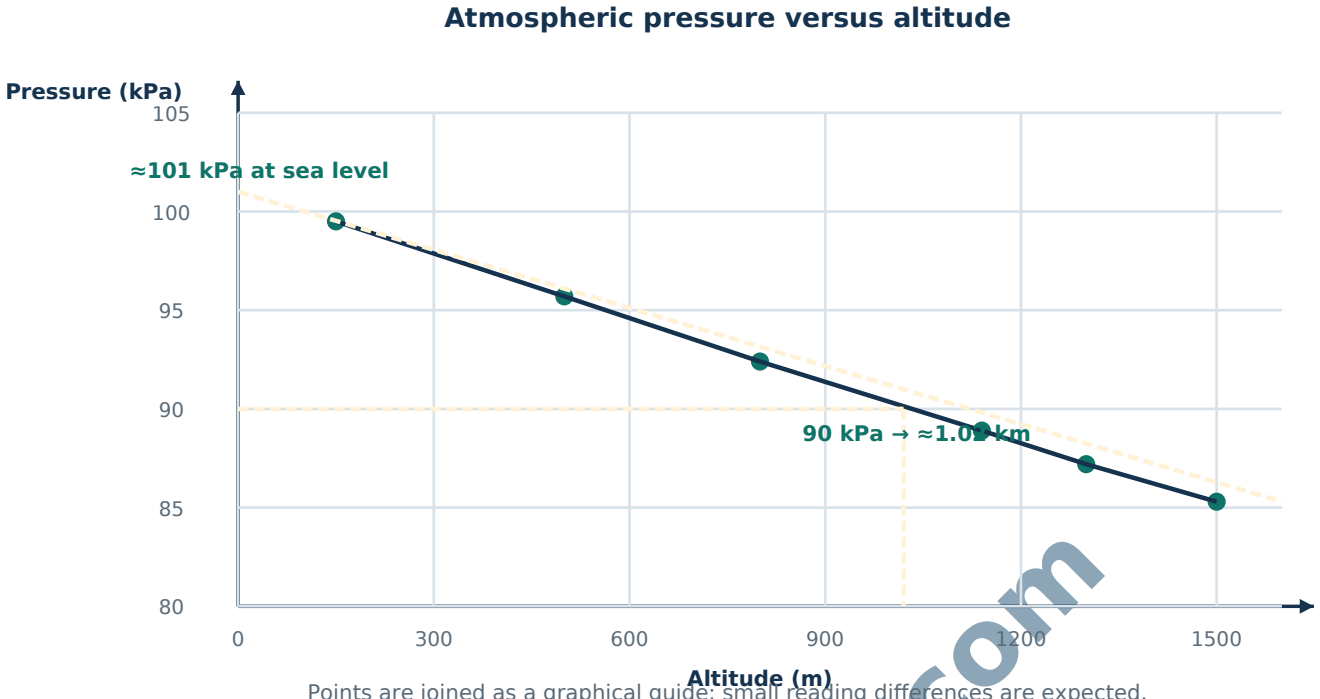
$$F_1 = F_2(A_1/A_2) = 4000 \times 10/100 = 400 \text{ N}$$

Answer: 400 N.

6.10. A hot-air balloon records the following data. Draw an altitude–pressure graph and estimate (a) sea-level pressure and (b) altitude where pressure is 90 kPa.

Altitude (m)	150	500	800	1140	1300	1500
Pressure (kPa)	99.5	95.7	92.4	88.9	87.2	85.3

Figure 3 — Plotted data with graphical extrapolation and 90 kPa read-off



Points are joined as a graphical guide; small reading differences are expected.

Extend the trend back to zero altitude: **(a) $P_0 \approx 101$ kPa = 1.01×10^5 Pa**. Read across from 90 kPa to the trend and down to the altitude axis: **(b) h ≈ 1.02 km**. Graph-reading values within a small tolerance are acceptable.

6.11 If pressure in a hydraulic press is increased by an additional 10 N cm⁻², how much extra load will the 50 cm² output platform support?

Given: $\Delta P = 10$ N cm^{-2}; $A = 50$ cm^2. Areas cancel in matching units.

$$\Delta F = \Delta P \times A = 10 \times 50 = 500 \text{ N}$$

Answer: Extra load = 500 N.

6.12 OR: A 500 N force acts normally on a hydraulic-brake piston of area 5 cm². Find (a) pressure transferred to the oil and (b) force on a 20 cm² brake piston.

Given: This official alternative repeats the data of E6.7: $A_1 = 5 \times 10^{-4}$ m^2; $A_2 = 20 \times 10^{-4}$ m^2.

$$\text{(a) } P = F_1/A_1 = 500/(5 \times 10^{-4}) = 1.0 \times 10^6 \text{ N m}^{-2}$$

$$\text{(b) } F_2 = F_1(A_2/A_1) = 500 \times 20/5 = 2000 \text{ N}$$

Answer: (a) 1.0×10^6 Pa (b) 2000 N.

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F. More Important Questions (from the text)

1. How do stress and strain describe extension beyond the simple spring equation?

Tensile stress is force per cross-sectional area, σ = F/A, measured in Pa. **Tensile strain** is fractional extension, ε = ΔL/L, and has no unit. Within the proportional region, stress ∝ strain.

Young modulus Y = stress/strain = (F/A)/(ΔL/L)

Therefore ΔL = FL/(AY)

This explains why a longer wire extends more, while a thicker wire extends less under the same load. For a fixed material and geometry, it reduces to Hooke's law F = kx.

2. What is the difference between gauge pressure and absolute pressure?

Absolute pressure = atmospheric (or surface) pressure + gauge pressure

For open liquid at depth h: $P_{abs} = P_{atm} + \rho gh$

Gauge pressure measures pressure relative to the surrounding atmosphere; a tyre gauge reads zero when the actual pressure inside equals atmospheric pressure. Absolute pressure is measured relative to vacuum and is used when total molecular pressure is required.

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Points to Remember

Hooke's law: F = kx within the elastic limit; k has unit N m^{-1}.

Springs: two in series → x_total = 2x and k_eff = k/2; parallel → x/2 and 2k.

Density: ρ = m/V; SI unit kg m^{-3}.

Pressure: P = F⊥/A; 1 Pa = 1 N m^{-2}.

Wire extension: ΔL = FL/(AY), so ΔL ∝ L/d^2 for fixed load and material.

Liquid gauge pressure: p = ρgh; independent of container shape and plate area.

Absolute pressure: P_atm + ρgh in an open liquid.

Pascal: pressure change is transmitted undiminished through a confined fluid.

Hydraulics: $F_1/A_1 = F_2/A_2$; larger output area multiplies force but reduces travel.

Barometer: P_atm = ρgh; standard mercury height about 760 mm.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	d	3	a	4	d	5	d
6	b	7	b	8	a				

How to attempt this chapter in the paper

- Convert cm^2 by multiplying by 10^{-4} and cm^3 by 10^{-6}; litres convert to m^3 by 10^{-3}.
- For maximum pressure choose minimum contact area; for minimum pressure choose maximum area.
- Do not add atmospheric pressure when the question asks only for water or gauge pressure.
- Hydraulic force ratios use areas, not diameters; diameter ratios must be squared.
- A barometer measures vertical mercury height, not the tube length.
- Elasticity is resistance to deformation and recovery, not merely the ability to stretch greatly.